# RELATIVE PSEUDO-COHERENCE TRANSFERS BETWEEN BASE AND INTERMEDIATE WHEN THE INTERMEDIATE STRUCTURE SHEAF IS PSEUDO-COHERENT

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*Editorial note.* Statement span is lines 17234-17246 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17248-17272, the complete author proof (23 body lines).

*Editorial note.* Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Composition law of the chapter's relative-pseudo-coherence family; the footnote's dictionary (O_Y pc over S means Y -> S pc) is retained.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. Definition of relative pseudo-coherence and pseudo-coherent morphisms (chapter definition section with the footnote dictionary); More on Algebra, lemma pull-relative-pseudo-coherent (the diagram technique the proof copies); chapter lemma base-change-relative-pseudo-coherent; Derived Categories of Schemes, lemma closed-push-pseudo-coherent.

*Source reference.* Original lemma 067Z: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17375–17381. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 09UR: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17112–17120. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 067B: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17272–17272. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 09VA: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17272–17272. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Date*: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

**Lemma 0.1.** *Let $f : X \to Y$ be a morphism of schemes locally of finite type over a base $S$. Let $m \in \mathbf{Z}$. Let $E$ be an object of $D(\mathcal{O}_X)$. Assume $\mathcal{O}_Y$ is pseudo-coherent relative to $S$[1]. Then the following are equivalent*

(1) *$E$ is $m$-pseudo-coherent relative to $Y$, and*
(2) *$E$ is $m$-pseudo-coherent relative to $S$.*

*Proof.* The question is local on $X$, hence we may assume $X$, $Y$, and $S$ are affine. Arguing as in the proof of More on Algebra, Lemma 067B we can find a commutative diagram

$$\begin{array}{ccccc} X & \xrightarrow{i} & \mathbf{A}^m_Y & \xrightarrow{j} & \mathbf{A}^{n+m}_S \\ {\scriptstyle f}\downarrow & \swarrow{\scriptstyle p} & & \swarrow & \\ Y & \longrightarrow & \mathbf{A}^n_S & & \end{array}$$

The assumption that $\mathcal{O}_Y$ is pseudo-coherent relative to $S$ implies that $\mathcal{O}_{\mathbf{A}^m_Y}$ is pseudo-coherent relative to $\mathbf{A}^m_S$ (by flat base change; this can be seen by using for example Lemma 09UR). This in turn implies that $j_*\mathcal{O}_{\mathbf{A}^n_Y}$ is pseudo-coherent as an $\mathcal{O}_{\mathbf{A}^{n+m}_S}$-module. Then the equivalence of the lemma follows from Derived Categories of Schemes, Lemma 09VA. □

[1]This means $Y \to S$ is pseudo-coherent, see Definition 067Z.